



EUROPEAN
SPALLATION
SOURCE



GUI approaches at ESS

Software for the science programme, from proposal to analysis

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European Spallation Source (ESS)

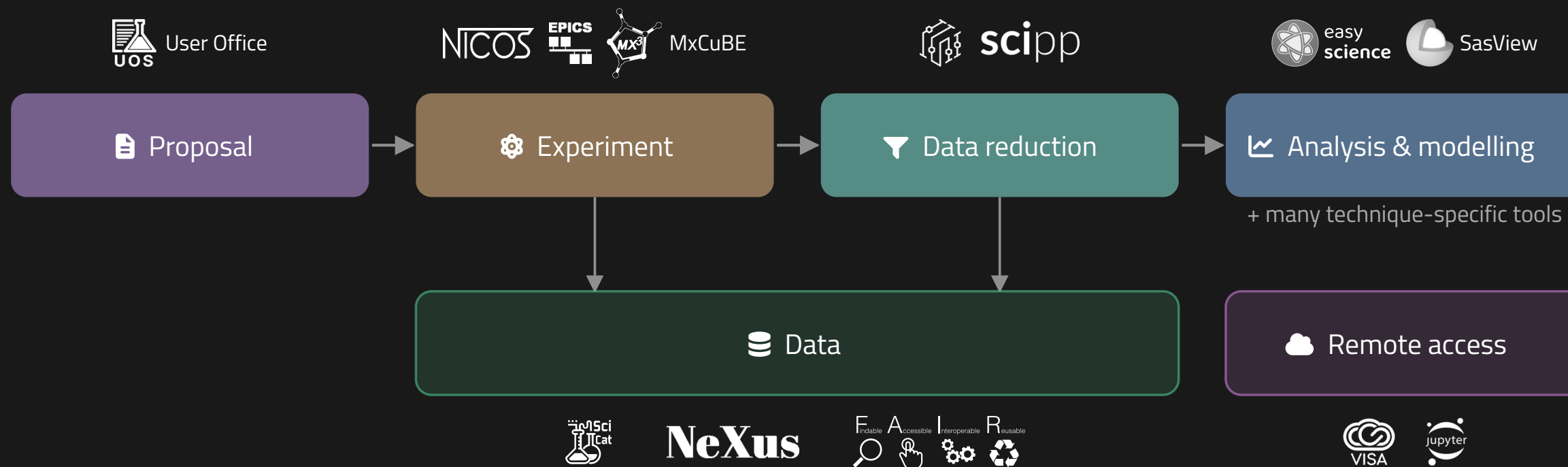
Data Management and Scientific Computing (DMSC)

NOBUGS 2026 · GUI Workshop · 21 Sep 2026

The integrated data pipeline

Different user interfaces support each stage of the data pipeline.

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UI technologies in a nutshell

Qt for desktop applications. Web tools for other stages.

DESKTOP

Qt Widgets · PyQt6

 **NICOS**

nicos-controls.org

 MLZ

- Beamline control GUI
- ESS uses custom panels
- One layout per instrument

Qt Widgets · PySide6

 **SasView**

sasview.org

 many facilities

- Small-angle scattering analysis
- A new toolkit means a rewrite
- Adopted with the project

Qt QML · PySide6

 **EasyDiffraction · EasyReflectometry · EasyImaging**

easyscience.org

 U Göttingen  JCNS at MLZ  Chalmers

- Declarative UI in QML, separate from Python code
- EasyDynamics, EasyShapes and EasyTexture prototypes
- Prototypes become products without a rewrite

WEB

ipywidgets · Voilà

 **essreduce**

scipp.github.io/ess/reduce

 STFC

- Generated from a scipp workflow
- Documented, not deployed at ESS
- Works inside notebooks

Streamlit · Python

SANS-webapp

github.com/ai4se1dk/SANS-webapp

 AI4SE1DK

- SANS fitting
- AI assistant built in
- Little GUI code to maintain

Angular · TypeScript

 **SciCat**

scicatproject.org

 PSI  MAX IV

- Data catalogue
- Find and access datasets
- Browsed from any institution

React · TypeScript

 **User Office**

github.com/UserOfficeProject

 STFC  ELI  Diamond

- Proposal submission
- Facility access
- Used before anyone arrives

Challenges and controls

Most challenges are **organisational**. Beamline and accelerator controls use different UI approaches.

Organisational

- Little is shared between groups
 - Each group picks its own stack
- GUI expertise stays inside each group
 - No common review or style practice
- The same people do backend and GUI work

Technical

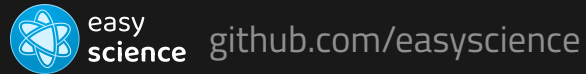
- Scientific plotting rarely supports QML
 - Embedding plots takes extra work
 - Embedded plots may not match the UI style
- Built-in QML charts lack some features
 - They would need work for scattering data
- ESS maintains a NICOS fork
 - We aim to converge with the main project

Beamline vs accelerator

- Beamline GUIs come from ECDC in DMSC
 - One shared NICOS client
 - Phoebus (Java desktop) for EPICS screens
- Accelerator GUIs come from elsewhere
 - Another division, another stack
 - No UI components in common

Reusable UI components

Shared QML components across EasyScience applications.



What we share

- The workflow bar, the panels and the controls
- Used by every EasyScience application
- **Shared only within EasyScience**
 - Other ESS projects use their own components

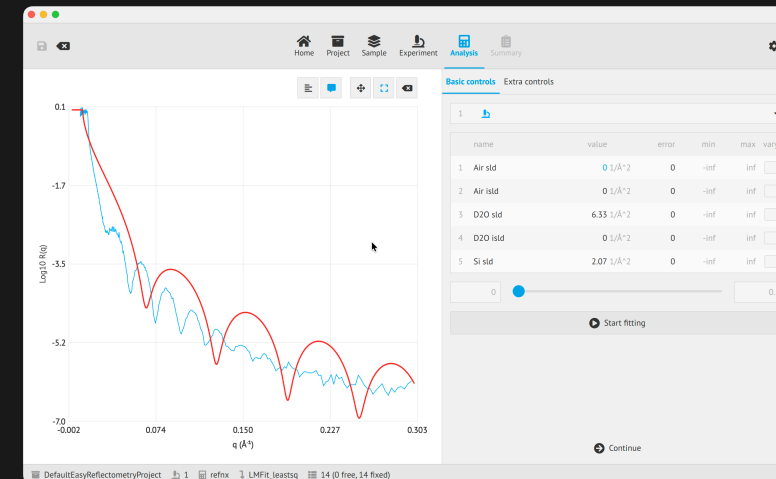
How we document it

- Five examples in the repository
- The applications are the main reference
 - The docs site is set up but not written
- **Next: write the docs and update them each release**

EasyDiffraction · diffraction data analysis



EasyReflectometry · reflectometry data analysis



UI generation and AI assistants

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An AI assistant is available today. Two UI generation ideas are not built yet.

✂ UI generation

💡 ideas

- Two directions we are considering
- We could generate data reduction notebooks for Voilà with AI
 - The workflow already describes the parameters
- We could assemble a new EasyScience GUI with AI
 - Using our shared components and existing applications as examples

🤖 AI assistants

SANS-webapp github.com/ai4se1dk/SANS-webapp

- Suggests a scattering model from the data
- Answers questions about the fit
- **Sets parameters and runs the fit when asked**
 - Its tools are off until you switch them on

SANS-webapp · AI-assisted model selection, with the chat panel on the right

